

Section 01

Industrial Security (Bachelor)



Section 01 – Introduction to Industrial Security

Introduction to Industrial Security

Lecture Notes

Department of Computer Science

- 1 Cyber-Physical Foundations
- 2 Context and History
- 3 Threat Setting and Misconceptions

By the end of this section you will be able to

- Define Operational Technology (OT) and contrast it with Information Technology (IT).
- Identify the core components of an Industrial Control System (ICS).
- Explain why the CIA triad is re-ordered for industrial environments.
- Place industrial security in its historical and economic context.
- Recognise common misconceptions about ICS security.

1

Cyber-Physical Foundations

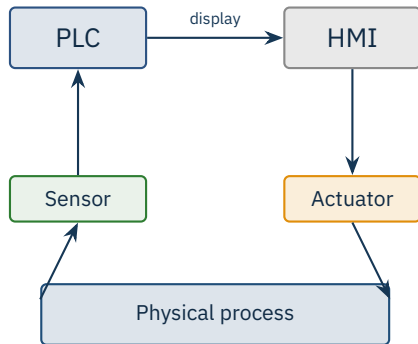
Section 01 – Introduction to Industrial Security

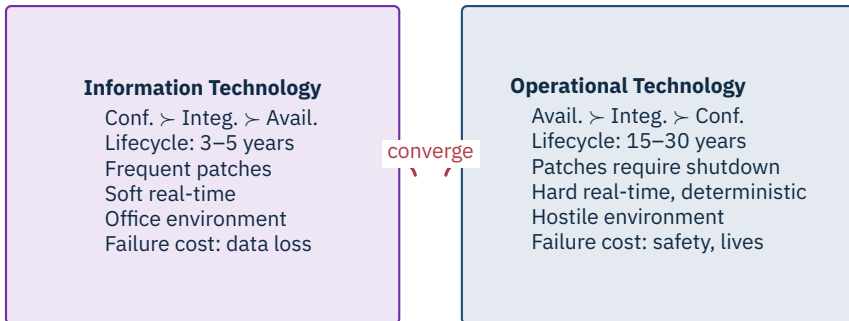
Definition

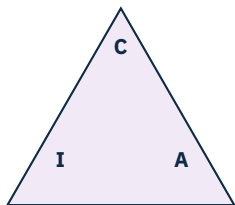
Industrial security is the protection of cyber-physical systems — computers that monitor and control machines in the physical world.

Sectors: power, water, oil & gas, manufacturing, transport, building automation.

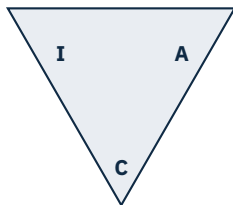
A breach has consequences in *atoms*, not just in *bits*.







IT priorities



OT priorities

Remember

In OT, **availability** and **safety** dominate the CIA triad. A blocked alarm can blind an operator to a runaway reactor.

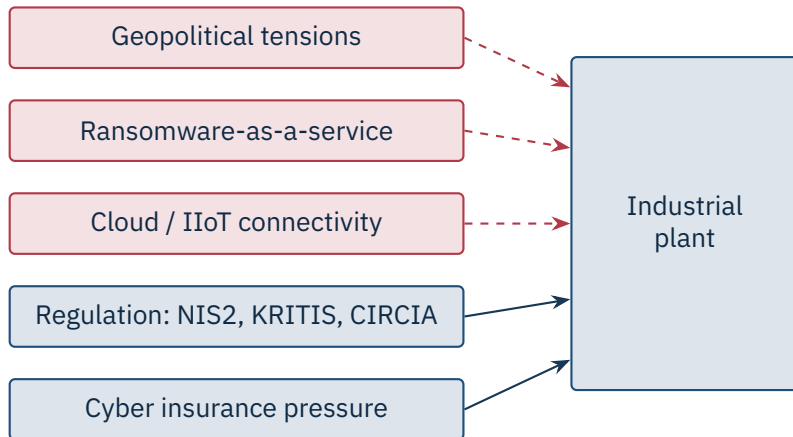
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Context and History

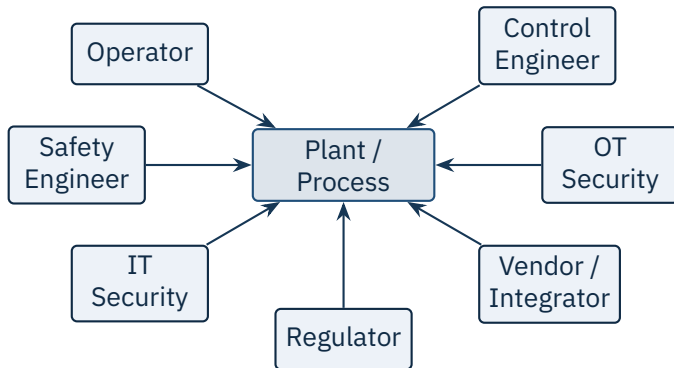
Section 01 – Introduction to Industrial Security



Each milestone made systems more capable – and more reachable.



Source: Dragos Year in Review 2024: 23 active threat groups tracked, ICS ransomware up 87% YoY | Munich Re Cyber Insurance Risk Snapshot, 2024.



Industrial security is a team sport across roles that rarely sat together historically.

3

Threat Setting and Misconceptions

Section 01 – Introduction to Industrial Security

“We’re air-gapped, so we’re safe.”

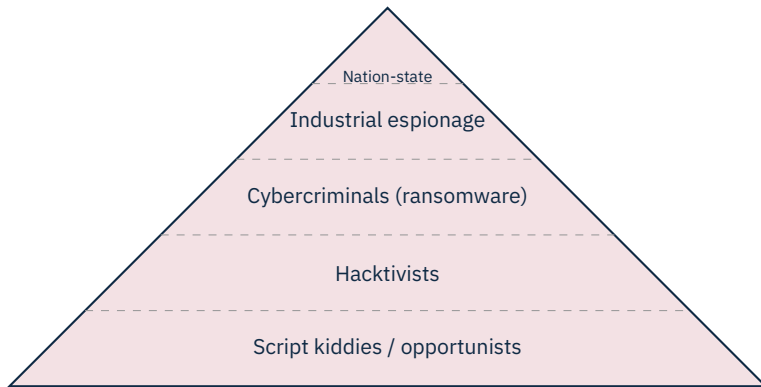
“Our protocols are obscure; nobody understands them.”

“We’ve never been attacked, so we’re fine.”

“IT tools will solve OT problems.”

“Patching solves everything.”

“Compliance equals security.”



Capability rises as we move up; volume of actors falls.

Colonial Pipeline

\$4.4M ransom
5-day outage

Norsk Hydro

\$71M loss
6-month recovery

Maersk (NotPetya)

\$300M loss
10 days down

Stuxnet

~1000 centrifuges
multi-year program

Direct + indirect

Recovery costs typically exceed ransom by 5–10×. The lasting damage is reputation and regulatory.

Source: Colonial: FBI 2021; Norsk Hydro Q1 2019 report; Maersk Q2 2017 interim report; Stuxnet centrifuge count: D. Albright et al., ISIS, 2010.



New attack surface

Every cloud connector is a new ingress – and a new authentication challenge for legacy field devices.

- **OT Security Engineer** – design, deploy, monitor controls; bridges IT and OT
- **Control Engineer with security focus** – starts from process side
- **ICS Penetration Tester** – offensive; specialised employers
- **Compliance / Audit** – IEC 62443, NIS2 specialist
- **Incident Responder (OT)** – on-call, often consulting-side (Dragos, Mandiant)
- **Researcher** – vendor / academic; vulnerabilities, threat intel

Demand

Headhunter reports consistently rank OT security in the top three hardest-to-fill cyber roles globally.

ICS & SCADA fundamentals
Network architectures
Industrial protocols
Threat landscape
Vulnerabilities & assessment
Risk management
Defense in depth
Standards & governance
Incident response

★ Key Takeaway: Industrial security is different from IT security

- OT differs fundamentally from IT in goals, lifecycle, and tolerance for failure.
- Availability and safety dominate the CIA triad in industrial contexts.
- Adversaries today span from script kiddies to nation-state APTs.
- The next sections build the toolbox: components, networks, protocols, defenses.

- Albright, D., Brannan, P., Walrond, C. *Did Stuxnet take out 1,000 centrifuges at the Natanz enrichment plant?* ISIS, 22 Dec 2010.
- Falliere, N., Murchu, L., Chien, E. *W32.Stuxnet Dossier*, v1.4. Symantec, Feb 2011.
- FBI. *Statement on the Colonial Pipeline attack*, May 2021.
- GAO. *Critical infrastructure protection: Federal authorities for the Colonial Pipeline response*, GAO-22-104746, 2022.
- Norsk Hydro. *Q1 2019 Interim Report* (cyber-attack cost disclosure).
- A.P. Moller – Maersk. *Q2 2017 Interim Report* (NotPetya cost disclosure).
- NIST. *Cybersecurity Framework 2.0*, Feb 2024.
- ENISA. *Threat Landscape*, annual.
- Knapp, E.D., Langill, J.T. *Industrial Network Security*, 2nd ed., Syngress, 2014.
- Bodungen, C.E. et al. *Hacking Exposed: Industrial Control Systems*, McGraw-Hill, 2016.

End of Section 01



Introduction to Industrial Security

Questions and discussion